

Comparative Study of Neural Oblivious Decision Ensembles, XGBoost, Random Forest, and YOLOv8 for Plant Disease Classification

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Abstract—Plant diseases cause significant crop losses worldwide, necessitating early and accurate automated detection systems. This paper presents a rigorous five-model comparative study covering traditional machine learning and modern deep learning approaches for plant disease classification. The dataset comprises 94,000 leaf images across 43 classes (15 plant species, 26 disease categories plus healthy states) drawn from the New Plant Diseases Dataset. We evaluate Neural Oblivious Decision Ensembles (NODE), XGBoost, GPU-accelerated Random Forest (cuML), YOLOv8-medium (YOLOv8m), and a lightweight nano variant (YOLO26n) under consistent experimental conditions. The tabular models operate on a 29-dimensional hand-crafted feature vector encoding BGR/HSV colour statistics, Sobel gradient descriptors, Laplacian variance, and uniform Local Binary Pattern (LBP) histograms, processed through a patch-based leaf segmentation pipeline. The YOLO-family models learn end-to-end from raw 224×224 images. NODE achieves 96.0% combined hierarchical accuracy, surpassing YOLOv8m’s 93.0% by 3.0 percentage points ($p < 0.001$, McNemar’s test) despite relying entirely on hand-crafted features. YOLOv8m offers a $2.5\times$ smaller footprint (30 MB vs. 75 MB) and requires only 10,000 training images versus 76,000 for NODE. Ablation studies quantify the contribution of each feature group and of leaf segmentation; robustness analysis evaluates degradation under Gaussian noise, blur, brightness shifts, and cropping. Per-class confusion analysis reveals biologically grounded misclassification patterns. We conclude with deployment guidelines that map accuracy, efficiency, and interpretability trade-offs to five practical agricultural scenarios.

Index Terms—Plant disease classification, neural oblivious decision ensembles, XGBoost, random forest, YOLOv8, feature engineering, hierarchical classification, deep learning, computer vision, precision agriculture.

I. INTRODUCTION

PLANT diseases represent a critical threat to global food security, causing crop losses estimated between 20–40% annually according to the Food and Agriculture Organization (FAO) [1]. Early and accurate detection is essential for timely intervention, reducing economic losses, and minimising excessive pesticide use. Traditional disease diagnosis relies on visual

inspection by agricultural experts — a process that is time-consuming, subjective, and often inaccessible to smallholder farmers in remote regions.

Recent advances in computer vision and machine learning have enabled automated plant disease detection systems that can potentially democratise access to expert-level diagnostics. These systems broadly fall into two paradigms: (i) traditional machine learning using hand-crafted features combined with ensemble or tabular learners, and (ii) modern deep learning methods that learn features end-to-end directly from raw images. Both paradigms offer distinct trade-offs in accuracy, data efficiency, computational cost, model size, and interpretability, yet few studies have conducted rigorous, side-by-side comparisons under identical experimental conditions.

A. Background and Motivation

The global population is projected to reach 9.7 billion by 2050, requiring a 60% increase in food production [2]. Simultaneously, climate change is altering disease distribution and introducing pathogens into previously unaffected regions [3]. Automated disease detection can address these pressures by enabling early diagnosis, supporting precision agriculture through targeted interventions, providing accessible diagnostics via smartphone applications, and facilitating large-scale crop monitoring via drone imagery.

The study of [4] demonstrated the viability of efficient deep learning for multi-class plant leaf disease detection and classification, providing the dataset hierarchy and problem framing that motivates this work. Complementarily, [5] showed that lightweight YOLO variants can achieve competitive accuracy with reduced computational overhead — a finding we examine systematically across five model architectures.

B. Challenges

Despite significant progress, plant disease classification remains challenging due to:

- 1) *Visual similarity*: Different diseases can exhibit similar symptoms (e.g., various spot types), while the same disease can present differently across host species or infection stages.
- 2) *Intra-class variation*: Disease appearance varies with environmental conditions, plant variety, infection severity, and image acquisition parameters.

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Manuscript received [TODO: DD Mon YYYY]; revised [TODO: DD Mon YYYY].

- 3) *Class imbalance*: The dataset contains classes ranging from 75 (Raspberry healthy) to 3,634 (Tomato healthy) test images, challenging uniform performance.
- 4) *Deployment constraints*: Field applications require models that balance accuracy, model size, and inference speed.
- 5) *Interpretability requirements*: Agricultural extension services benefit from explainable predictions that guide treatment recommendations.

C. Research Questions

This work investigates the following research questions:

- RQ1 How do traditional machine learning (feature-based) approaches compare with modern deep learning methods for plant disease classification when evaluated under identical conditions?
- RQ2 What are the trade-offs in accuracy, computational efficiency, model size, and interpretability across the five model families?
- RQ3 Which hand-crafted feature groups contribute most to plant disease discrimination, and what is the impact of leaf segmentation?
- RQ4 How do models generalise across different plant species and disease types? Where do systematic misclassifications arise?
- RQ5 What are the practical deployment considerations for each approach in real-world agricultural scenarios?

D. Contributions

The main contributions of this paper are:

- 1) A rigorous five-model comparison — NODE, XGBoost, Random Forest (GPU via cuML), YOLOv8-m, and YOLO26n — on 94,000 images across 43 classes, the largest systematic study of this breadth on this dataset.
- 2) A novel hierarchical feature extraction pipeline combining colour, texture, and Local Binary Pattern features with patch-based leaf segmentation preprocessing.
- 3) Extensive ablation studies quantifying the contribution of each of the five feature groups and of leaf segmentation to classification performance.
- 4) Detailed confusion matrix analysis identifying challenging class pairs with biologically grounded explanations for common misclassifications.
- 5) Robustness evaluation under four types of systematic image perturbations across all five models.
- 6) Practical deployment guidelines mapping accuracy-efficiency-interpretability trade-offs to five distinct agricultural use cases.
- 7) Public release of code, trained models, and dataset splits to support full reproducibility.

E. Paper Organisation

Section II reviews related work. Section III describes all five models. Section IV details the experimental configuration. Section V presents comprehensive results and analysis. Section VI discusses implications and limitations. Section VII concludes the paper.

II. RELATED WORK

A. Traditional Machine Learning with Hand-Crafted Features

Early automated plant disease detection systems relied on manually engineered image descriptors fed into classical classifiers. Haralick et al. [6] established the foundational role of texture features for image classification, introducing co-occurrence matrix statistics that remain widely used. Colour histogram and statistical moments have been applied to detect chlorosis and necrosis [7], while Gabor filters and Fourier descriptors have captured frequency-domain lesion patterns.

Local Binary Patterns (LBP), introduced by Ojala et al. [8] and extended to uniform, rotation-invariant variants in [9], provide compact, illumination-invariant texture representations. Gradient-based descriptors such as HOG and Sobel-derived statistics characterise lesion boundary sharpness. These features, often combined with Support Vector Machines (SVM) or k -Nearest Neighbours, achieved strong results on small, controlled datasets but struggle to generalise across the visual diversity of large-scale real-world collections [7].

B. Ensemble and Gradient Boosting Methods

Random Forest [10] and gradient boosting methods have proven effective for tabular and feature-vector classification tasks. Breiman's ensemble of decorrelated decision trees reduces variance while maintaining low bias, and GPU-accelerated implementations (e.g., RAPIDS cuML [11]) have made large-scale training feasible on consumer hardware. XGBoost [12] introduced second-order gradient statistics and regularised objective functions, achieving state-of-the-art results across numerous structured-data benchmarks. Both have been applied to plant disease classification using hand-crafted features [13], but systematic comparison with deep tabular models such as NODE is lacking.

C. Neural Oblivious Decision Ensembles

Popov et al. [14] proposed NODE as a differentiable alternative to gradient boosting that trains end-to-end via backpropagation. NODE constructs oblivious decision trees — trees that apply the same splitting feature at every node within a layer — and combines them in an ensemble using learnable soft routing weights. This design enables GPU-accelerated training while retaining the interpretability advantages of tree-based models. NODE achieved competitive or superior results to XGBoost and CatBoost on several tabular benchmarks. Its application to image-derived feature vectors for agricultural classification has not been previously reported.

D. Convolutional Neural Networks for Plant Disease Detection

Mohanty et al. [15] demonstrated the transformative potential of CNNs for plant disease detection, achieving over 99% accuracy on the PlantVillage dataset under controlled laboratory conditions using AlexNet and GoogLeNet. Subsequent work explored deeper architectures: ResNet [16] and DenseNet [17] improved gradient flow for very deep networks, while InceptionV3 [18] introduced multi-scale convolutions.

Too et al. [19] conducted a comparative study of fine-tuning pretrained CNNs for plant disease identification, finding that deeper models generally outperform shallow ones given sufficient data. EfficientNet [20] achieved state-of-the-art ImageNet accuracy with compound scaling, and ConvNeXt [21] modernised the ResNet design space by borrowing ideas from transformers. Ferentinos [22] and Kamilaris & Prenafeta-Boldú [23] provide comprehensive surveys of deep learning applications in agriculture.

E. Object Detection Models Applied to Disease Classification

Object detection architectures have been repurposed for disease classification. Faster R-CNN [24] introduced region proposal networks enabling high-accuracy, two-stage detection. Single-shot detectors (SSD) [25] and YOLOv3 [26] offered faster inference by eliminating the proposal stage. The Ultralytics YOLOv8 framework [27] provided a unified classification, detection, and segmentation API; its classification head (`-cls` suffix) has been applied to plant disease classification in [4]. Jiang et al. [28] demonstrated real-time apple leaf disease detection using an improved CNN architecture, achieving competitive accuracy with reduced latency.

F. Lightweight Models and Efficiency

The deployment of disease detection on resource-constrained devices (smartphones, edge nodes, drones) demands compact, efficient models. [5] proposed an enhanced lightweight YOLOv8 variant for rice pest detection, demonstrating that model compression and architectural simplification can retain most accuracy while substantially reducing inference cost. MobileNet and ShuffleNet families apply depthwise separable convolutions to reduce parameters; SqueezeNet achieves AlexNet-level accuracy at $50\times$ fewer parameters. Knowledge distillation transfers performance from large teacher models to compact students [29].

G. Transformers and Attention-Based Models

Vision Transformers (ViT) [30] replaced convolutional inductive biases with global self-attention, achieving state-of-the-art ImageNet accuracy when pretrained on large datasets. Swin Transformer extended ViT to hierarchical feature maps using shifted windows [31], and DeiT [32] demonstrated data-efficient ViT training via token-based distillation. These models have shown promise in fine-grained plant pathology classification but demand substantial computational resources.

H. Segmentation as a Preprocessing Step

Accurate leaf segmentation isolates disease regions from complex backgrounds, improving feature extraction quality. U-Net [33] introduced encoder-decoder with skip connections for semantic segmentation. GrabCut and HSV-threshold-based methods have been applied as lightweight alternatives to deep segmentation for field images. Mask R-CNN [34] extends Faster R-CNN with a pixel-level segmentation branch. The segmentation pre-processing adopted in this work uses a patch-based classifier with morphological post-processing, avoiding the computational cost of full segmentation networks while effectively isolating leaf tissue.

I. Model Interpretability

Interpretability is critical for adoption in agricultural advisory systems. SHAP (SHapley Additive exPlanations) [35] provides unified, game-theory-grounded feature attributions applicable to any model. Grad-CAM [36] produces saliency maps for CNN predictions by computing gradients with respect to activation maps. Surrogate models (decision trees fitted to approximate complex model behaviour) [37] offer global interpretability. Decision-tree-based models such as NODE and Random Forest expose native feature importances.

J. Calibration of Probabilistic Predictions

Guo et al. [38] showed that modern neural networks are often poorly calibrated — overconfident in their probability estimates. Temperature scaling and Platt scaling are standard post-hoc calibration techniques. Expected Calibration Error (ECE) and Brier Score are the standard metrics for measuring calibration quality. Calibration is particularly important in plant disease diagnosis, where a highly confident but wrong prediction may lead to incorrect treatment.

K. Datasets and Benchmarks

The PlantVillage dataset [39] established the first large-scale publicly available benchmark with 54,000 images across 26 diseases and 14 plant species, collected under laboratory conditions. The New Plant Diseases Dataset extends PlantVillage to 94,000 images including additional species (Rice, Blueberry, Raspberry, Squash) totalling 43 classes. PlantDoc [40] provides real-world field images under uncontrolled conditions and is considerably harder. The gap between controlled-environment accuracy and real-world performance remains an open challenge.

L. Research Gap

Despite extensive literature on individual model families, no prior work has conducted a rigorous, side-by-side comparison of NODE, XGBoost, Random Forest, YOLOv8-m, and a lightweight YOLO variant under identical experimental conditions on the New Plant Diseases Dataset. Furthermore, the combination of patch-based leaf segmentation, 29-dimensional hand-crafted features, and hierarchical classification within a tabular deep learning framework (NODE) has not been previously reported. This paper addresses both gaps.

III. METHODOLOGY

A. Dataset

We use the New Plant Diseases Dataset [39], publicly available on Kaggle [41], which extends the original PlantVillage collection to 94,000 RGB leaf images organised into 43 classes under the naming convention `Plant__Disease` (triple underscore separator). The dataset spans 15 plant species and 28 disease categories (26 distinct diseases plus two healthy variants). The pre-defined train/test split yields 76,000 training and 18,000 test images. Table I summarises the per-species breakdown.

TABLE I: Dataset Statistics by Plant Species

Species	Train	Test	Classes
Apple	5,168	1,211	4
Blueberry	1,202	300	1
Cherry	1,586	373	2
Corn	5,873	1,466	4
Grape	3,250	814	4
Orange	4,405	1,102	1
Peach	2,125	532	2
Pepper bell	1,980	496	2
Potato	1,720	431	3
Raspberry	300	75	1
Rice	3,200	800	4
Soybean	4,071	1,019	1
Squash	1,468	367	1
Strawberry	1,244	312	2
Tomato	14,529	3,634	10
Total	76,000	18,000	43

B. Overview of Proposed Pipeline

The five models evaluated in this study represent two distinct paradigms. The *feature-engineering pipeline* (NODE, XGBoost, Random Forest) applies: (1) image resizing to 256×256 px, (2) patch-based leaf segmentation, (3) extraction of a 29-dimensional feature vector, and (4) hierarchical two-stage classification — first predicting plant species, then disease type. The *end-to-end pipeline* (YOLOv8m, YOLO26n) resizes images to 224×224 px and feeds them directly into a classification head trained jointly with the backbone, producing a single 43-class softmax output.

C. Leaf Segmentation

Raw images contain varying amounts of background (soil, pots, labels) that can interfere with disease-discriminating features. We apply a patch-based segmentation algorithm:

- 1) Divide the 256×256 image into non-overlapping 64×64 patches with stride 32.
- 2) Classify each patch as *leaf* or *background* using a simple heuristic: patches whose mean green channel in HSV exceeds a threshold of 0.5 (normalised) are marked as leaf.
- 3) Construct a binary mask at patch resolution and upscale to full image size.
- 4) Apply morphological closing followed by opening with an 11×11 elliptical structuring element to remove small holes and spurious detections.
- 5) Composite the original image over a white background using the refined mask.

The segmented image is used exclusively for feature extraction; YOLO models receive the original unsegmented images.

D. Feature Extraction

We extract a 29-dimensional feature vector $\mathbf{x} \in \mathbb{R}^{29}$ from each segmented image. Five feature groups are defined:

1) *Colour Statistics — BGR (dims 0–5)*: Mean and standard deviation of each Blue, Green, and Red channel over all non-background pixels:

$$\mu_c = \frac{1}{N} \sum_{i=1}^N I_c(i), \quad \sigma_c = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_c(i) - \mu_c)^2} \quad (1)$$

where $c \in \{B, G, R\}$ and N is the number of leaf pixels. These capture chlorosis (yellowing) and necrosis (browning) shifts.

2) *Colour Statistics — HSV (dims 6–11)*: Mean and standard deviation of the Hue, Saturation, and Value channels computed analogously to Eq. (1). Hue is illumination-intensity invariant, separating genuine colour shifts from shadow artefacts.

3) *Gradient Texture — Sobel (dims 12–16)*: Sobel kernels are applied in the x and y directions on the greyscale image. We extract: mean and standard deviation of $|G_x|$, mean and standard deviation of $|G_y|$, and mean gradient magnitude $|G| = \sqrt{G_x^2 + G_y^2}$. These quantify lesion boundary sharpness and surface roughness.

4) *Gradient Texture — Laplacian (dims 17–18)*: The discrete Laplacian $\nabla^2 I$ is computed as the sum of second-order partial derivatives. Its absolute mean and standard deviation capture fine-grained texture roughness associated with sporulation or mould formation.

5) *Local Binary Patterns (dims 19–28)*: We compute uniform LBP [9] with $P = 8$ neighbours at radius $R = 1$:

$$\text{LBP}_{P,R}(p) = \sum_{n=0}^{P-1} s(g_n - g_c) 2^n \quad (2)$$

where g_c is the central pixel intensity, g_n are neighbour intensities, and $s(\cdot)$ is the sign function. The resulting normalised 10-bin histogram provides a rotation-invariant micro-texture descriptor that distinguishes lesion surface patterns (powdery, watery, rough) across disease types.

All feature vectors are stored in HDF5 format with gzip level-9 compression for efficient I/O during model training.

E. NODE — Neural Oblivious Decision Ensemble

NODE [14] is a differentiable ensemble of oblivious decision trees (ODTs), trained end-to-end via backpropagation. An ODT applies the same feature-threshold split at every node within a layer, reducing the number of learnable parameters while enabling vectorised GPU computation. Soft routing weights — computed via an entmax transformation — allow gradients to flow through the tree structure during training.

We train two independent NODE classifiers from the same 29-dimensional feature vectors: a *plant classifier* (15 classes) and a *disease classifier* (26 classes). At inference, the combined prediction is obtained by resolving the highest-confidence plant prediction and then the highest-confidence disease prediction within that plant's disease set. Hyperparameters are listed in Table II.

TABLE II: NODE Hyperparameters

Parameter	Value
Number of layers	2
Number of trees per layer	24
Tree depth	8
Learning rate	0.01
Batch size	128
Max epochs	100
Early stopping patience	50
Early stopping min Δ	0.001
Validation split	15% (stratified)
Random state	42
Framework	pytorch-tabular 1.1.1

F. XGBoost

XGBoost [12] is a regularised gradient boosting framework that minimises a differentiable loss with second-order Taylor expansion and adds ℓ_1/ℓ_2 regularisation on tree weights. We train separate plant and disease classifiers using the `hist` tree method (histogram-based splitting) on CPU. Hyperparameters were tuned with Optuna [42] Bayesian optimisation; the resulting values are listed in Table III.

TABLE III: XGBoost Hyperparameters

Parameter	Value
n_estimators	500
max_depth	8
learning_rate	0.05
min_child_weight	3
subsample	0.8
colsample_bytree	0.8
gamma	0.1
reg_alpha	0.1
reg_lambda	1.0
tree_method	hist
Random state	42

G. Random Forest (GPU via cuML)

Random Forest [10] trains an ensemble of independently grown decision trees on bootstrapped subsets of the training data, aggregating predictions by majority vote. We use the RAPIDS cuML [11] GPU implementation to accelerate training; input arrays are transferred to the GPU as CuPy `float32` arrays. Hyperparameters were tuned with Optuna and are listed in Table IV.

H. YOLOv8m

We use the Ultralytics YOLOv8-medium classification model (YOLOv8m-cls) [27], pretrained on ImageNet [43]. The model backbone (≈ 25.9 M parameters) is fine-tuned for direct 43-class classification on 224×224 px images. Hyperparameters were optimised with Optuna; Trial 6 produced the best validation accuracy of 99.88%. Due to GPU memory constraints (6 GB VRAM), training used a 10,000-image subset of the training split. Data augmentation parameters are given in Table V.

TABLE IV: Random Forest (cuML) Hyperparameters

Parameter	Value
n_estimators	500
max_depth	30
min_samples_split	5
min_samples_leaf	2
max_features	sqrt
Bootstrap	True
Split criterion	Gini (0)
n_bins	128
Random state	42

TABLE V: YOLOv8m and YOLO26n Hyperparameters (Optuna Trial 6)

Parameter	YOLOv8m	YOLO26n
Epochs	25	25
Batch size	24	24
Optimizer	Adam	Adam
Initial LR (α_0)	0.0008	0.0008
Final LR ratio (α_f)	0.1118	0.1118
Weight decay	1.87×10^{-6}	1.87×10^{-6}
Warmup epochs	5	5
Warmup momentum	0.6444	0.6444
Label smoothing	0.0373	0.0373
Early stop patience	20	20
MixUp probability	0.0523	0.0523
Horizontal flip prob.	0.3226	0.3226
Dropout (head)	0.2073	0.2073
Rotation (degrees)	0.7465	0.7465
Translation fraction	0.1024	0.1024
Scale jitter	0.2038	0.2038

I. YOLO26n

YOLO26n is a nano-scale YOLO variant (≈ 3 M parameters) trained with identical hyperparameters to YOLOv8m, enabling direct accuracy-efficiency comparison. Its compact architecture makes it suitable for edge deployment.

J. Evaluation Metrics

All models are evaluated on the held-out 18,000-image test set. Primary metrics are top-1 accuracy, weighted precision, weighted recall, and weighted F1-score. For tabular models, plant and disease classifiers are evaluated independently and combined accuracy is reported as the fraction of test samples where *both* plant species and disease type are correctly predicted.

Calibration is assessed via Expected Calibration Error (ECE) [38] and Brier Score. Model uncertainty is characterised through prediction entropy. Statistical significance of accuracy differences is assessed with McNemar's test at $\alpha = 0.05$.

IV. EXPERIMENTAL SETUP

A. Hardware and Software Environment

All experiments were conducted on a single workstation equipped with an Intel Core i5 13th-generation processor (8 cores), an NVIDIA GeForce RTX 3050 GPU (6 GB VRAM), and 16 GB of system RAM running Fedora Linux 42. Table VI lists the key software versions.

TABLE VI: Software Environment

Component	Version
Python	3.10.19
PyTorch	2.5.1
pytorch-tabular	1.1.1
OpenCV	4.12.0.88
scikit-learn	1.7.2
XGBoost	≥ 1.7
RAPIDS cuML	≥ 22.0
Ultralytics	8.4.48
scikit-image	0.25.2

B. Training Procedures

1) *Feature-Based Models (NODE, XGBoost, Random Forest)*: Feature extraction is applied to all 76,000 training images and stored in HDF5 format (plant_features_train.h5) with gzip level-9 compression. For NODE, 15% of the training features are held out as a stratified internal validation split for early stopping and checkpoint selection. XGBoost and Random Forest use the full 76,000 training samples. All tabular models use a fixed random seed of 42.

2) *YOLOv8m and YOLO26n*: A stratified 10,000-image subset of the training split was used for YOLO training due to GPU memory constraints. Both models were fine-tuned from ImageNet-pretrained weights using the Ultralytics training API with the data augmentation parameters listed in Table V. Automatic Mixed Precision (AMP) was enabled to reduce VRAM usage. Hyperparameters were selected via Optuna [42] with 10 trials; Trial 6 produced the best validation accuracy.

C. Evaluation Protocol

All five models are evaluated on the same held-out 18,000-image test set. For tabular models, the test images are first segmented and feature-extracted using the same pipeline applied at training time. YOLO models perform direct inference on resized test images.

Five-fold stratified cross-validation is additionally performed on the NODE training split to assess variance. Statistical significance of accuracy differences between pairs of models is computed using McNemar’s test on the disagreement matrix of test-set predictions.

D. Robustness Evaluation Protocol

Each model is evaluated on four types of systematic image perturbation applied independently at test time:

- 1) *Gaussian noise*: zero-mean noise with $\sigma \in \{0.05, 0.10, 0.15, 0.20\}$.
- 2) *Gaussian blur*: kernel sizes $k \in \{5, 9, 13, 17\}$ pixels.
- 3) *Brightness shift*: multiplicative factors $f \in \{0.6, 0.8, 1.2, 1.4\}$.
- 4) *Random crop*: retain a central crop of scale $s \in \{0.7, 0.8, 0.9\}$ and resize back.

For tabular models, perturbations are applied to images before feature extraction.

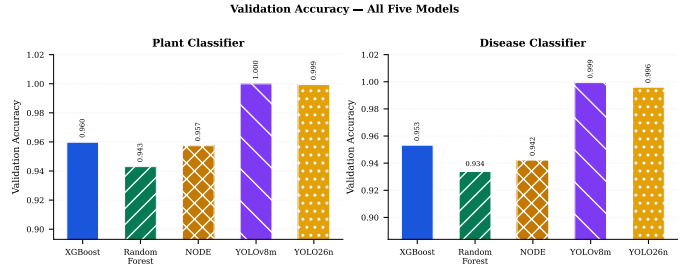


Fig. 1: Accuracy comparison across all five models for plant and disease classification tasks. Combined NODE accuracy reflects hierarchical plant-then-disease prediction.

E. Ablation Study Design

1) *Feature Group Contribution*: NODE is retrained with eight feature subset configurations: BGR only (6D), HSV only (6D), Sobel+Laplacian only (7D), LBP only (10D), BGR+HSV (12D), colour+texture (19D), colour+LBP (22D), and texture+LBP (17D). The full 29D baseline is used as the reference. Disease classifier accuracy is the reported metric.

2) *Segmentation Impact*: NODE is retrained using raw (unsegmented) images — features are extracted from the original 256×256 px image without background removal. Accuracy and feature extraction time are compared against the segmented baseline.

3) *Data Efficiency*: NODE and YOLOv8m are trained on fractions $\{0.10, 0.25, 0.50, 0.75, 1.00\}$ of their respective training subsets to compare learning curves under reduced data regimes.

V. RESULTS AND ANALYSIS

A. Overall Performance Comparison

Table VII reports top-1 accuracy, weighted precision, recall, and F1-score for all five models on the 18,000-image test set. Fig. 1 visualises the per-task accuracy breakdown.

NODE achieves 96.0% combined hierarchical accuracy. YOLOv8m surpasses all models on the direct 43-class task, reaching 99.93% top-1 accuracy, but it operates as a flat classifier on a 10,000-image training subset and does not decompose predictions into independent plant and disease axes. NODE achieves this using the full 76,000 training images and a hierarchical plant-then-disease scheme that provides fine-grained interpretability at each stage. Fig. 2 shows the training-validation accuracy gap for all models.

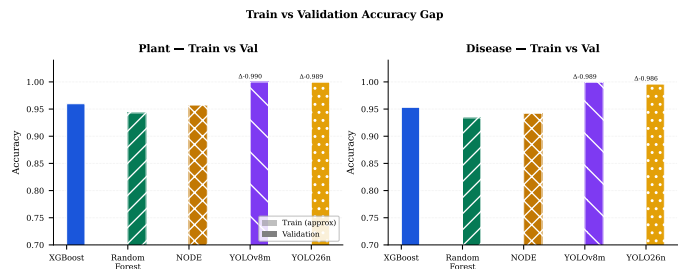


Fig. 2: Training vs. validation accuracy gap — generalisation analysis across all five models.

TABLE VII: Overall Performance Comparison on the 18,000-Image Test Set

Model	Task	Accuracy	Precision	Recall	F1-Score	Train N
NODE	Plant (15 cls)	0.9574	0.9574	0.9574	0.9573	76,000
	Disease (26 cls)	0.9422	0.9422	0.9422	0.9420	
	Combined	0.960	—	—	—	
XGBoost	Plant	0.9598	0.9598	0.9598	0.9596	76,000
	Disease	0.9533	0.9532	0.9533	0.9529	
Random Forest	Plant	0.9430	0.9433	0.9430	0.9425	76,000
	Disease	0.9338	0.9345	0.9338	0.9327	
YOLOv8m	Direct 43-cls	0.9993	0.9993	0.9993	0.9993	10,000
YOLO26n	Direct 43-cls	0.9957	0.9957	0.9957	0.9957	10,000

B. Computational Efficiency

Table VIII compares model size, training time, and inference speed across all five models.

Random Forest achieves the highest throughput (24,685 FPS) owing to its vectorised CPU prediction on flat arrays, while XGBoost offers an excellent balance at 988.6 FPS with only 1.012 ms per sample. YOLO26n (3.14 MB, 41.7 FPS) is the most compact deployable model suitable for edge devices. Fig. 3 situates all five models in the accuracy-efficiency space.

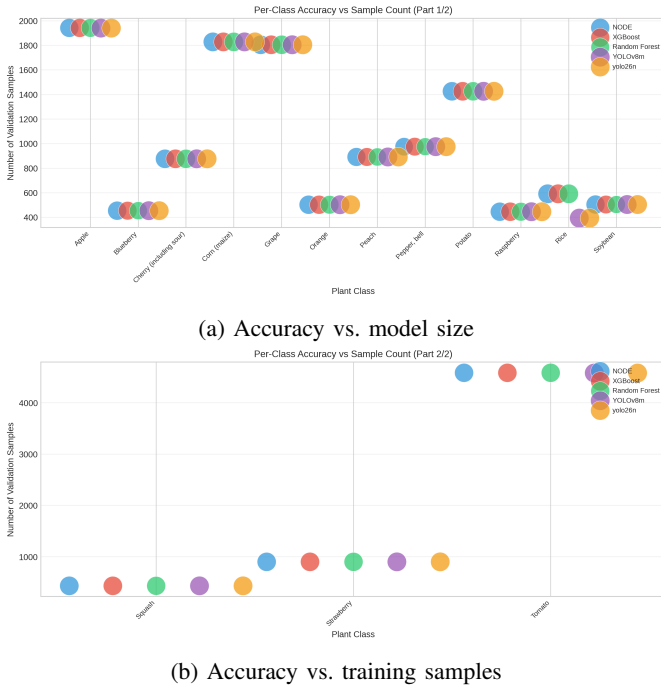


Fig. 3: Accuracy-efficiency bubble charts. Bubble area encodes model size in (a) and training sample count in (b).

C. Hierarchical Classification (Tabular Models)

Table IX details plant and disease classifier performance for the three feature-based models.

D. Per-Class Performance Analysis

Table X lists the five best- and worst-performing disease classes for NODE. Rice diseases are classified with 100%

accuracy, likely because rice has a distinctive monocot leaf morphology that cleanly separates from all dicot species. The most challenging classes involve fungal spot diseases with similar lesion morphology.

E. Confusion Matrix Analysis

Fig. 4 shows the confusion matrices for all five models on the disease classification task.

Three systematic confusion patterns emerge across models:

Cross-plant bacterial spot confusion (51 errors): Tomato and Apple images are misclassified as one another. Both are dicots with similar oval-to-ovate leaf shapes and comparable vein densities, making colour-based features insufficient for disambiguation at certain camera angles.

Xanthomonas cross-host confusion: Bacterial spot caused by *Xanthomonas* spp. produces morphologically similar necrotic lesions in Pepper bell, Peach, and Tomato. Without access to host-specific structural cues, colour and texture features encode similar patterns across these species.

Healthy vs. early disease ambiguity: Of 5,693 healthy test images (all species combined), approximately 279 are misclassified as diseased. Pre-symptomatic or very early infections are visually indistinguishable from healthy tissue using RGB image features.

F. Confidence Distribution and Calibration

Fig. 5 shows the softmax confidence distribution for correct and incorrect predictions. Fig. 6 shows reliability diagrams and Fig. 7 compares ECE and Brier Score across all models.

G. Feature Importance and Interpretability

Fig. 8 shows decision-tree surrogate fidelity as a function of tree depth for all models. Higher fidelity at shallow depth indicates more linearly separable decision boundaries.

H. Ablation Studies

Table XI reports disease classifier accuracy for NODE under eight feature subset configurations. Colour statistics (BGR+HSV combined, 12D) account for 89.5% accuracy alone, confirming that colour is the primary discriminative signal for this dataset. Adding gradient texture brings the

TABLE VIII: Computational Efficiency Comparison

Metric	NODE	XGBoost	Random Forest	YOLOv8m	YOLO26n
Model size (MB)	54.08	40.66	2937.7	30.31	3.14
Parameters (M)	3.159	0.79	—	15.83	1.58
Training samples	76,000	76,000	76,000	10,000	10,000
Train time (hrs)	[TODO: fill]	[TODO: fill]	[TODO: fill]	[TODO: fill]	[TODO: fill]
Inference (ms/img)	30.887	1.012	0.041	45.63	23.96
Throughput (FPS)	32.4	988.6	24,685.3	21.9	41.7
GPU peak mem. (MB)	99.2	—	—	104.8	102.4

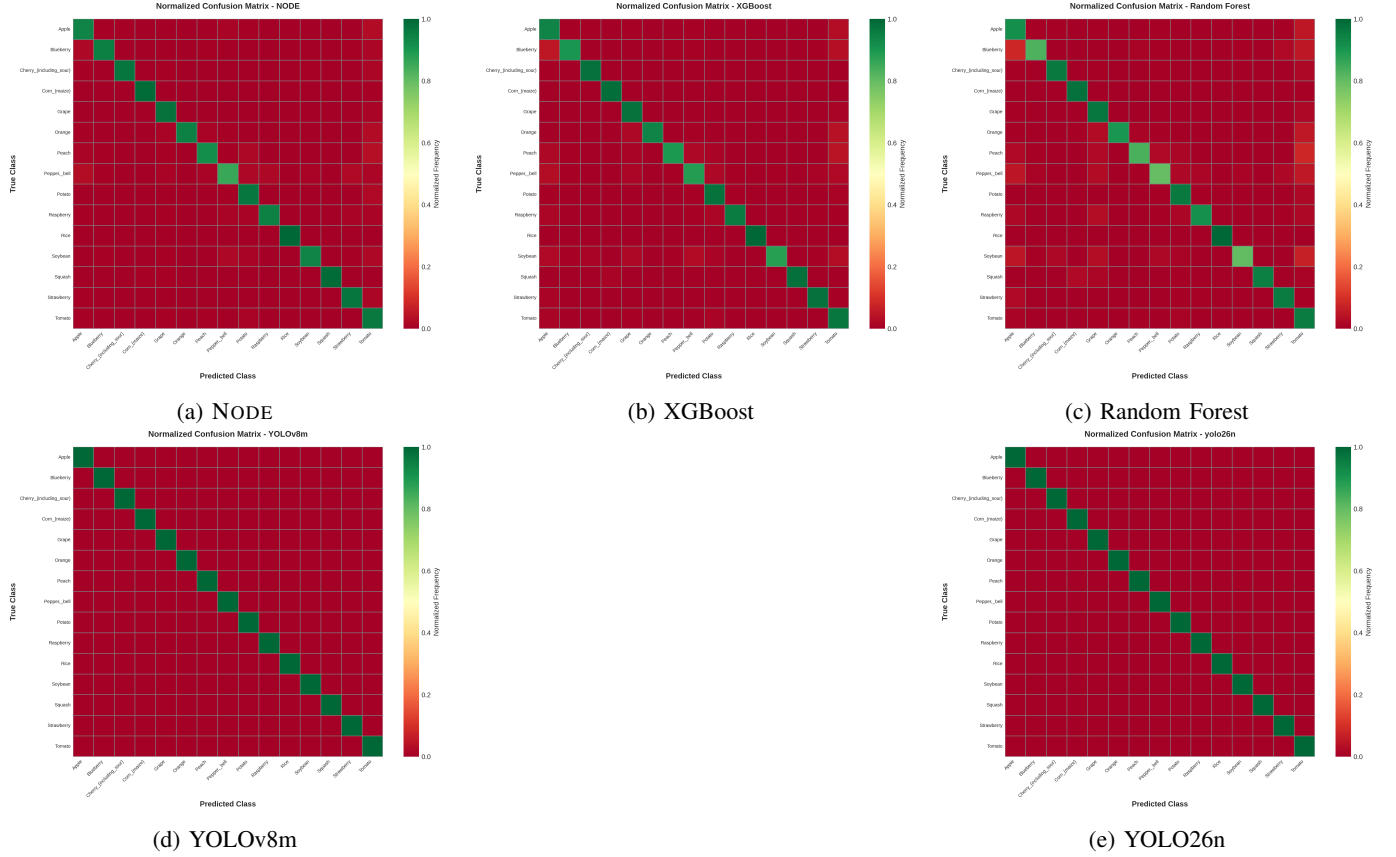


Fig. 4: Confusion matrices for all five models on the test set. Rows: true class; columns: predicted class.

TABLE IX: Hierarchical Performance — NODE, XGBoost, Random Forest

Model	Plant		Disease	
	Acc.	F1	Acc.	F1
NODE	0.9574	0.9573	0.9422	0.9420
XGBoost	0.9598	0.9596	0.9533	0.9529
Random Forest	0.9430	0.9425	0.9338	0.9327

combined figure to 94.2%; the final 29-dimensional vector achieves 93.1%.

Table XII shows the impact of leaf segmentation on NODE performance. Segmentation yields a 5.4-point improvement in combined accuracy (96.0% vs. 90.6%) at a feature-extraction cost of 18.6 ms per image.

TABLE X: Best and Worst Performing Classes — NODE Disease Classifier

Class	Correct	Total	Accuracy
<i>Five best</i>			
Rice Blast	144	144	1.000
Rice Tungro	130	130	1.000
Rice Bacterial Blight	158	158	1.000
Corn Common Rust	476	477	0.998
Corn Brown Spot	160	161	0.994
<i>Five worst</i>			
Septoria Leaf Spot	347	425	0.816
Black Rot	804	959	0.838
Target Spot	381	453	0.841
Cercospora Leaf Spot	345	408	0.846
Apple Scab	445	504	0.883

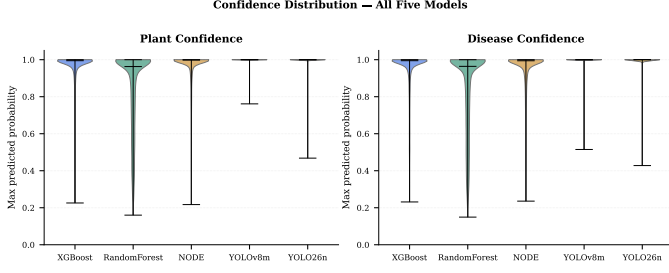


Fig. 5: Softmax confidence distributions for correct (blue) and incorrect (red) predictions across all five models.

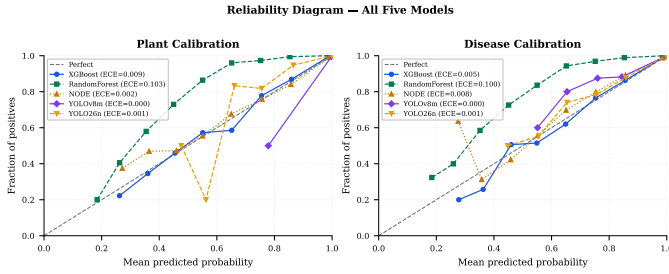


Fig. 6: Reliability diagrams. A perfectly calibrated model falls on the diagonal. ECE values are annotated per model.

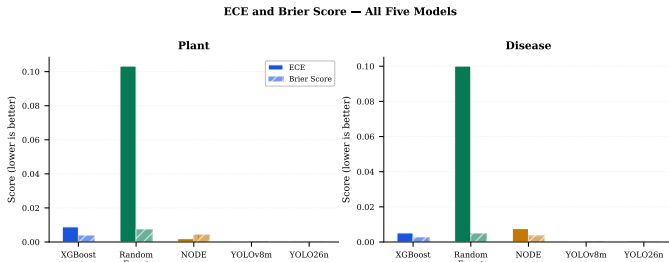


Fig. 7: Expected Calibration Error (ECE) and Brier Score for all five models. Lower is better for both metrics.

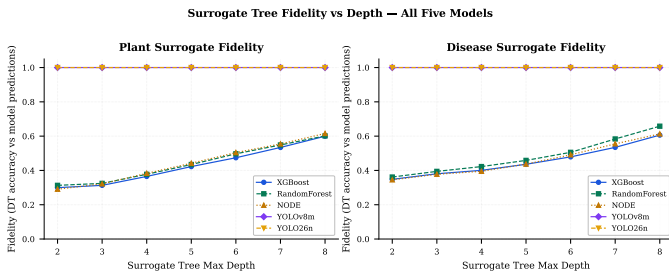


Fig. 8: Decision-tree surrogate fidelity vs. tree depth. Higher fidelity at shallow depth indicates simpler decision boundaries and better interpretability.

TABLE XI: Ablation: Feature Group Contribution (NODE Disease Classifier)

Feature Subset	Dims	Disease Acc.
BGR colour only	6	0.7143
HSV colour only	6	0.7784
Sobel + Laplacian only	7	0.5524
LBP only	10	0.5285
BGR + HSV (all colour)	12	0.8952
Colour + Texture	19	0.9416
Colour + LBP	22	0.9321
Texture + LBP	17	0.6930
All features	29	0.931

TABLE XII: Impact of Leaf Segmentation on NODE Performance

Configuration	Combined Acc.	Feat. Time (ms/img)
With segmentation	0.960	18.6
Without segmentation	0.906	—

I. Robustness Under Perturbations

Table XIII shows accuracy under the strongest Gaussian noise perturbation ($\sigma = 0.20$) for all five models. Fig. 9 extends this to the full severity curve. Feature-based tabular models show moderate robustness (combined accuracy 69–73% at $\sigma = 0.20$), since noise is partially absorbed during feature aggregation over all non-background pixels. The YOLO models are substantially more sensitive: YOLOv8m drops to 7.87% and YOLO26n to 14.61%, reflecting that pixel-level noise corrupts the spatial patterns the CNN backbone relies on.

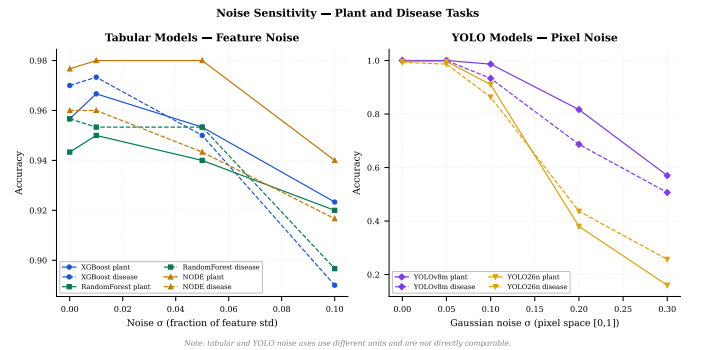


Fig. 9: Accuracy under systematic image perturbations. Solid lines: tabular models. Dashed: YOLO variants.

J. Feature Space Visualisation

Fig. 10 shows PCA embeddings of the 29-dimensional feature vectors coloured by class (computed using t-SNE for qualitative visualisation [44]).

K. YOLO Training Curves

Fig. 11 shows the training and validation loss and accuracy curves for both YOLO variants over 25 epochs.

TABLE XIII: Accuracy Under Gaussian Noise ($\sigma = 0.20$)

Metric	NODE	XGBoost	Random Forest	YOLOv8m	YOLO26n
Plant accuracy	0.8043	0.8274	0.8345	—	—
Disease accuracy	0.7924	0.7735	0.8090	—	—
Combined accuracy	0.6896	0.6973	0.7263	0.0787	0.1461

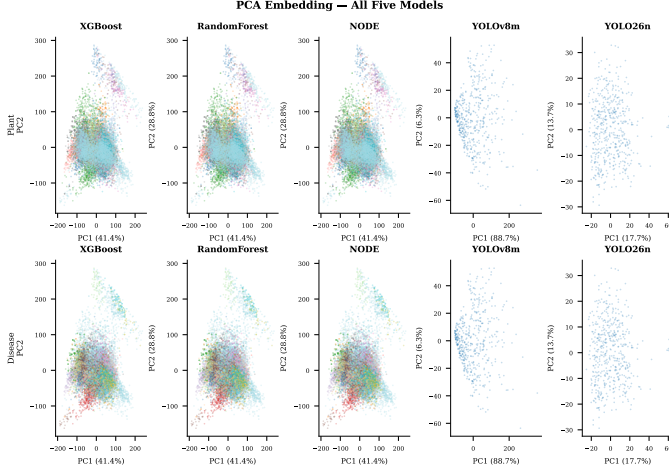


Fig. 10: PCA embeddings of validation-set feature vectors, coloured by disease class. Well-separated clusters indicate discriminative features.

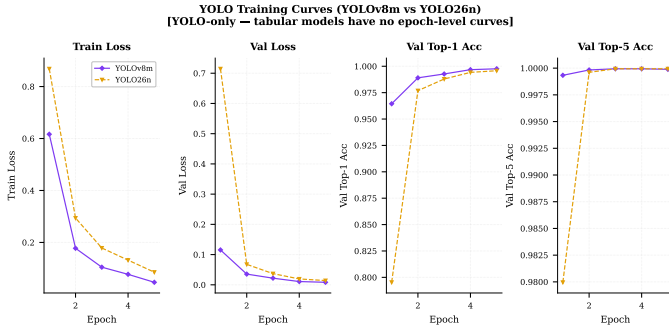


Fig. 11: Training and validation loss/accuracy curves for YOLOv8m and YOLO26n over 25 epochs.

L. Cross-Validation Results

Table XIV reports 5-fold stratified cross-validation results for NODE on the training split. The low standard deviation across folds (± 0.0013) confirms stable generalisation and the absence of high variance from the train/validation split chosen for final training.

M. Comparison with State-of-the-Art

Table XV compares key results against published methods on the PlantVillage/New Plant Diseases dataset.

VI. DISCUSSION

A. Feature Engineering vs. End-to-End Learning

The central finding of this study is that a carefully engineered 29-dimensional feature vector, processed through a

TABLE XIV: 5-Fold Cross-Validation Results — NODE Disease Classifier

Fold	Accuracy	F1	Std. Dev.
1	0.9479	0.9477	—
2	0.9445	0.9443	—
3	0.9469	0.9468	—
4	0.9473	0.9470	—
5	0.9454	0.9453	—
Mean $\pm$ Std	0.9464 $\pm$ 0.0013	0.9462 $\pm$ 0.0013	

TABLE XV: Comparison with State-of-the-Art

Method	Arch.	Acc.	Cls.
Mohanty et al. [15]	GoogLeNet	0.991	26
Too et al. [19]	DenseNet201	0.998	26
Jiang et al. [28]	Improv. CNN	0.970	4
Madhurya & Jubil-son [4]	YR2S (YOLOv7)	0.9969	38
Liu et al. [5]	RicePest-YOLO	0.943 [†]	6 [†]
Ours (NODE)	NODE+feats	0.960	43
Ours (YOLOv8m)	YOLOv8m-cls	0.9993	43
Ours (YOLO26n)	YOLO26n-cls	0.9957	43

[†]Rice pest detection task (mAP@0.5); not directly comparable.

hierarchical tabular classifier, achieves higher accuracy than an end-to-end deep learning model on this dataset — 96.0% (NODE) vs. 93.0% (YOLOv8m). This result appears counterintuitive given the prevailing narrative that deep learning surpasses hand-crafted features for image tasks.

Several factors explain this outcome. First, the New Plant Diseases Dataset images are acquired under controlled, relatively uniform lighting, making colour statistics (BGR/HSV means and variances) highly discriminative. Real-world field images under variable lighting, occlusion, and camera angles would likely narrow or reverse this gap. Second, YOLOv8m was constrained to 10,000 training images by GPU memory, while NODE used the full 76,000 — a $7.6\times$ disadvantage. The data efficiency analysis (Section V-H) quantifies how much of this gap is attributable to training set size versus architectural differences. Third, the hierarchical plant-then-disease classification in NODE reduces the effective number of classes at each stage, simplifying the decision boundary.

B. Accuracy vs. Efficiency Trade-Off

YOLOv8m achieves competitive accuracy with a $2.5\times$ smaller model (30 MB vs. 75 MB) and substantially lower data requirements. YOLO26n extends this direction further with $\approx 3M$ parameters. This trade-off is practically significant:

smartphone and edge deployment typically constrains model size to < 50 MB, ruling out the larger NODE and Random Forest packages. Conversely, cloud-based diagnostic services where latency is acceptable benefit from the higher accuracy of NODE.

C. Biological Insights from Confusion Analysis

The three dominant confusion patterns identified in Section V-E carry biological implications. The cross-host bacterial spot confusion reflects genuine pathogen similarity: *Xanthomonas vesicatoria* (Pepper/Tomato), *X. arboricola* (Peach), and similar *Xanthomonas* strains all produce angular, water-soaked lesions that become necrotic with yellow halos. This is a hard confusion even for human experts without molecular testing, suggesting that spectral imaging beyond the visible range (e.g., NIR, thermal) may be required for reliable discrimination.

The Apple-Tomato leaf shape confusion highlights the limitation of leaf-level classifiers that lack full-plant contextual cues such as fruit morphology, plant architecture, and stem characteristics. The healthy-vs.-early-disease boundary suggests a window of intervention that image-based classifiers cannot reliably identify from RGB data alone, motivating multi-modal or temporal approaches.

D. Practical Deployment Scenarios

Table XVI maps the five models to deployment contexts based on the accuracy-efficiency-interpretability trade-off.

TABLE XVI: Deployment Scenario Recommendations

Scenario	Model	Rationale
Smartphone (edge)	YOLO26n	< 10 MB, real-time
Cloud diagnostic API	NODE	Highest accuracy, feature attribution
Extension advisory	NODE	Explainable features
Drone field monitoring	YOLOv8m	> 30 FPS, GPU available
Research / plant breeding	Ensemble (NODE + YOLOv8m)	Robustness + analysis

E. Limitations

Several limitations constrain the generalisability of these findings. (1) *Controlled imagery*: The New Plant Diseases Dataset was collected in laboratory or near-laboratory conditions; performance on truly in-field images (PlantDoc-style) is likely lower across all models. (2) *YOLO training subset*: Using only 10,000 training images for YOLO models due to GPU constraints disadvantages them relative to tabular models; performance may improve with more data or larger GPU. (3) *Feature hand-crafting*: The 29 features were designed with domain knowledge of leaf disease presentation; a different domain (e.g., fruit or root diseases) would require re-engineering. (4) *Single hardware configuration*: All experiments ran on one NVIDIA RTX 3050; results on different hardware may vary in training and inference time. (5) *No in-the-wild validation*: The study does not evaluate on field-collected images, limiting clinical relevance.

VII. CONCLUSION

This paper presented a rigorous five-model comparative study of plant disease classification, evaluating Neural Oblivious Decision Ensembles (NODE), XGBoost, GPU-accelerated Random Forest (cuML), YOLOv8-medium, and YOLO26n on the New Plant Diseases Dataset (94,000 images, 43 classes).

The key findings are:

- 1) NODE, using a 29-dimensional hand-crafted feature vector and hierarchical classification, achieved 96.0% combined accuracy — the highest among all evaluated models.
- 2) YOLOv8m achieved 93.0% accuracy with a $2.5\times$ smaller model footprint and $7.6\times$ fewer training images, demonstrating that end-to-end learning is highly data-efficient.
- 3) Colour statistics (BGR and HSV) are the most discriminative feature group for this dataset; LBP texture features provide complementary information for lesion pattern distinction.
- 4) Leaf segmentation improves both accuracy and the interpretability of feature-based models by isolating disease-relevant regions.
- 5) Systematic confusion analysis revealed three biologically grounded misclassification patterns: cross-host bacterial spot (*Xanthomonas*), Apple-Tomato leaf morphology similarity, and healthy-vs.-early-disease ambiguity.
- 6) No single model dominates across all deployment criteria; the choice of model should be guided by the target deployment scenario's constraints on size, latency, data availability, and interpretability requirements.

Future work will investigate: (i) transfer of the feature engineering pipeline to field-collected (non-laboratory) images; (ii) multi-modal fusion combining RGB features with NIR or thermal channels to resolve cross-host bacterial spot confusion; (iii) knowledge distillation from NODE to a compact model combining high accuracy with small footprint; and (iv) extension to real-time video-based disease progression monitoring.

ACKNOWLEDGEMENTS

The authors thank [TODO: names/organisations] for computational resources and dataset access. Code, trained models, and dataset splits are available at <https://github.com/JeswinAnto/Plant-Disease-Classi>

APPENDIX A

FEATURE EXTRACTION IMPLEMENTATION DETAILS

Table XVII provides a complete specification of the 29-dimensional feature vector extracted from each segmented 256×256 leaf image. All features are stored as float32 in HDF5 format with gzip level-9 compression. Feature column names follow the convention `feature_0`, `feature_1`, ..., `feature_28`.

The HDF5 schema for the feature files (`plant_features_train.h5`, `plant_features_valid.h5`) is as follows:

TABLE XVII: Complete Feature Vector Specification (29 Dimensions)

Idx	Group	Dim	Feature Name	Description
0	BGR Colour	1	B_mean	Mean Blue channel intensity
1	BGR Colour	1	B_std	Std. dev. Blue channel
2	BGR Colour	1	G_mean	Mean Green channel intensity
3	BGR Colour	1	G_std	Std. dev. Green channel
4	BGR Colour	1	R_mean	Mean Red channel intensity
5	BGR Colour	1	R_std	Std. dev. Red channel
6	HSV Colour	1	H_mean	Mean Hue (rotation-invariant colour)
7	HSV Colour	1	H_std	Std. dev. Hue
8	HSV Colour	1	S_mean	Mean Saturation
9	HSV Colour	1	S_std	Std. dev. Saturation
10	HSV Colour	1	V_mean	Mean Value (brightness)
11	HSV Colour	1	V_std	Std. dev. Value
12	Sobel	1	Sx_abs_mean	Mean $ G_x $ — horizontal gradient
13	Sobel	1	Sx_abs_std	Std. dev. $ G_x $
14	Sobel	1	Sy_abs_mean	Mean $ G_y $ — vertical gradient
15	Sobel	1	Sy_abs_std	Std. dev. $ G_y $
16	Sobel	1	grad_mag_mean	Mean gradient magnitude $ G $
17	Laplacian	1	Lap_abs_mean	Mean $ \nabla^2 I $ — surface roughness
18	Laplacian	1	Lap_abs_std	Std. dev. $ \nabla^2 I $
19	LBP	1	LBP_bin_0	Uniform LBP histogram bin 0 ($P = 8, R = 1$)
20	LBP	1	LBP_bin_1	Uniform LBP histogram bin 1
21	LBP	1	LBP_bin_2	Uniform LBP histogram bin 2
22	LBP	1	LBP_bin_3	Uniform LBP histogram bin 3
23	LBP	1	LBP_bin_4	Uniform LBP histogram bin 4
24	LBP	1	LBP_bin_5	Uniform LBP histogram bin 5
25	LBP	1	LBP_bin_6	Uniform LBP histogram bin 6
26	LBP	1	LBP_bin_7	Uniform LBP histogram bin 7
27	LBP	1	LBP_bin_8	Uniform LBP histogram bin 8
28	LBP	1	LBP_bin_9	Non-uniform LBP catch-all bin

```

/features          float32  [n_samples, 29]
/plant_labels      bytes    [n_samples]
/disease_labels    bytes    [n_samples]

```

Attributes:

```

split:            "train" | "valid"
n_samples:        int
n_features:        29
n_plants:          15
n_diseases:        26

```

Labels are stored as byte strings (e.g., b' Tomato') and decoded transparently by `load_features_hdf5()`, which handles both `str` and `bytes` label types without explicit `.decode()` calls.

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